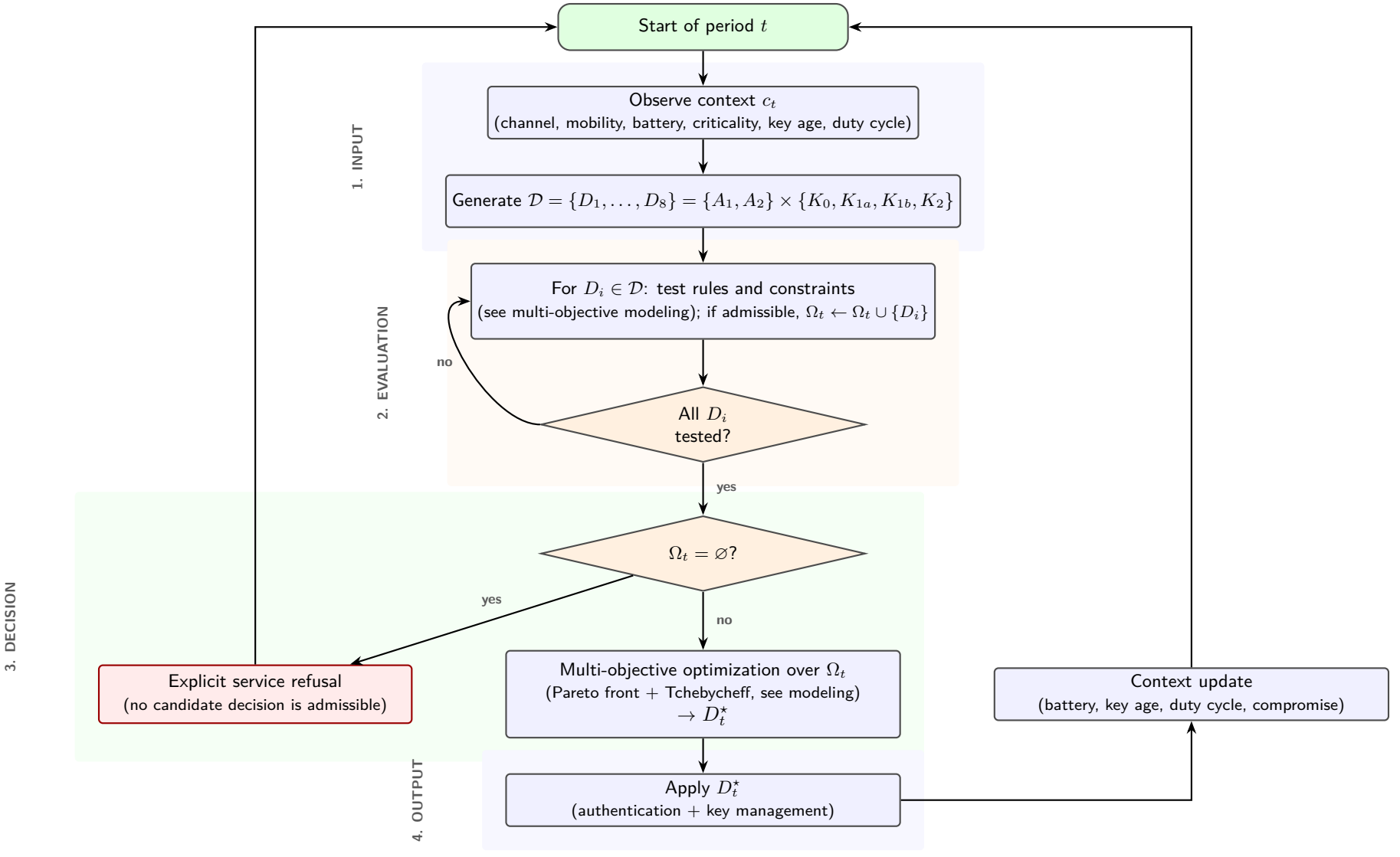
Start of period t
1. INPUT
Observe context c_t
(channel, mobility, battery, criticality, key age, duty cycle)
Generate $\mathcal{D} = \{D_1, \ldots, D_8\} = \{A_1, A_2\} \times \{K_0, K_{1a}, K_{1b}, K_2\}$
2. EVALUATION
For $D_i \in \mathcal{D}$: test rules and constraints
(see multi-objective modeling); if admissible, $\Omega_t \leftarrow \Omega_t \cup \{D_i\}$
no
All D_i tested?
yes
3. DECISION
$\Omega_t = \varnothing$?
yes
no
Explicit service refusal
(no candidate decision is admissible)
Multi-objective optimization over Ω_t
(Pareto front + Tchebycheff, see modeling)
$\rightarrow D_t^\star$
4. OUTPUT
Apply $D_t^\star$
(authentication + key management)
Context update
(battery, key age, duty cycle, compromise)